

YOUTRICOT TECHNICAL WHITEPAPER

Knitwear Tech Pack Guide

From Design Brief to Factory-Ready Specification

A practical specification framework for measurements, yarn, gauge, construction, packaging and factory handoff.

YouTricot

For fashion brands, product developers and sourcing teams



A tech pack is the operating contract for product development

When knitwear decisions remain implicit, factories must guess. A factory-ready specification converts design intent into measurable production inputs.

MEASURE

Fit becomes testable

Points of measure, grade rules and tolerances define the approval target.

ENGINEER

Materials become executable

Yarn, gauge, stitch density and construction are linked to the intended hand feel.

CONTROL

Handoffs become auditable

Revision history, sample comments and packaging instructions stay in one controlled file.

Generic apparel tech packs fail at knitwear-specific decisions

A silhouette sketch alone cannot control stretch, recovery, stitch density, linking, washing or finished garment dimensions.

Incomplete input

- Yarn count without supplier reference
- Gauge named without target density
- Measurements copied from cut-and-sew
- No wash or relaxation condition

Factory consequence

- Ambiguous hand feel and cost
- Unexpected weight and opacity
- Fit shifts after sampling
- Bulk variance after finishing

Seven blocks make a knitwear tech pack factory-ready

The file should guide commercial quoting, sampling, quality approval and bulk production without relying on undocumented assumptions.

01

Style identity

02

Measurements

03

Yarn & color

04

Gauge & density

05

Construction

06

Labels & packing

07

Revision control

Define how, where and when the garment is measured

Pair every point of measure with a method, base size and tolerance. Values below are illustrative and require style-specific validation.

POM	Measurement method	Size M	Tolerance
A	1/2 chest below armhole	52.0 cm	± 1.0 cm
B	HPS to bottom hem	60.0 cm	± 1.0 cm
C	Shoulder seam to cuff	58.0 cm	± 1.0 cm
D	Shoulder, seam to seam	40.0 cm	± 0.8 cm
E	Bottom opening, relaxed	44.0 cm	± 1.0 cm

Measure flat and relaxed, 24 hours after finishing, without stretching.

A yarn line must describe performance, not only composition

Premium yarn sourcing requires enough information for the factory to identify an equivalent, price it correctly and control shade consistency.

FIBER

90% merino wool / 10% cashmere

Quote and approval field

COUNT

2/26 Nm, two ends

Quote and approval field

SUPPLIER

Approved mill and article code

Quote and approval field

COLOR

Pantone reference + lab dip

Quote and approval field

PERFORMANCE

Pilling, shrinkage and colorfastness

Quote and approval field

HAND FEEL

Soft, compact surface; controlled halo

Quote and approval field

Gauge is a starting point; density controls the result

Machine gauge must be paired with yarn count, ply, stitch structure, target weight and washed density. Never approve gauge in isolation.

Gauge	Typical direction	Buyer check
3□?GG	Chunky structures; heavier yarn	Weight, seam bulk, recovery
7□?GG	Mid-gauge commercial sweaters	Opacity, drape, stitch definition
12GG	Fine-gauge merino and blends	Density, spirality, pilling
14□?6GG	Lightweight premium knitwear	Yarn evenness, snag risk, transparency

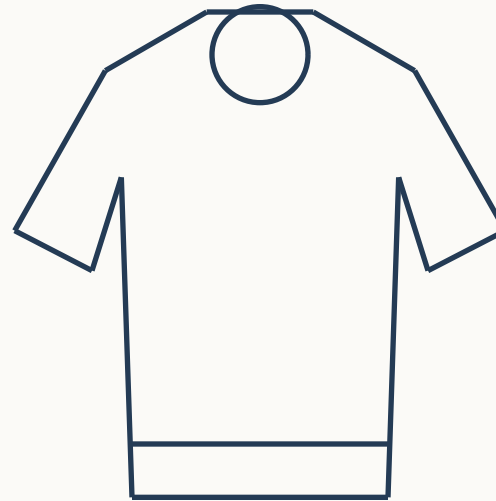
Factory confirmation: record courses and wales per 10 cm after washing, plus finished garment weight.

Show front, back and critical detail callouts

A clean technical drawing should identify stitch placement, neckline construction, rib dimensions, shoulder treatment, linking method and any engineered fashioning marks.



FRONT



BACK

DETAIL CALLOUTS

- 1x1 rib: 5 cm
- Fully fashioned shoulder
- Linking: 12GG
- Cable panel centered
- Tubular neckline finish

Translate design language into production instructions

Every decorative or structural element needs a location, dimension, stitch reference and approval method.

Stitch map

Name each body, sleeve, rib and trim structure.

Linking

Specify seam type, seam allowance and acceptable bulk.

Fashioning

Mark decreases, increases and visible needle marks.

Finishing

Define wash, steaming, brushing and pressing limits.

Approval

Attach reference swatch or approved sample photo.

Control every component that changes by color

A color page should connect yarn shade, contrast placement, labels, buttons, embroidery and packaging so the factory can build an accurate bill of materials.

NAVY

19-3921 TCX

OATMEAL

12-0605 TCX

MOSS

17-0613 TCX

- Approve lab dips under the agreed light source.
- List trim supplier, size, finish and attachment method.
- Use a component matrix for every colorway.

Packaging belongs in the tech pack, not in a late email

Clear packing instructions reduce relabeling, carton variance and delivery delays. Confirm buyer-specific legal information before production.

GARMENT

Fold method, tissue, hanger or flat pack

Record artwork version and placement

LABEL

Main, size, composition, care and country of origin

Record artwork version and placement

BAG

Material, warning text, size and ventilation

Record artwork version and placement

CARTON

Assortment, quantity, dimensions and weight

Record artwork version and placement

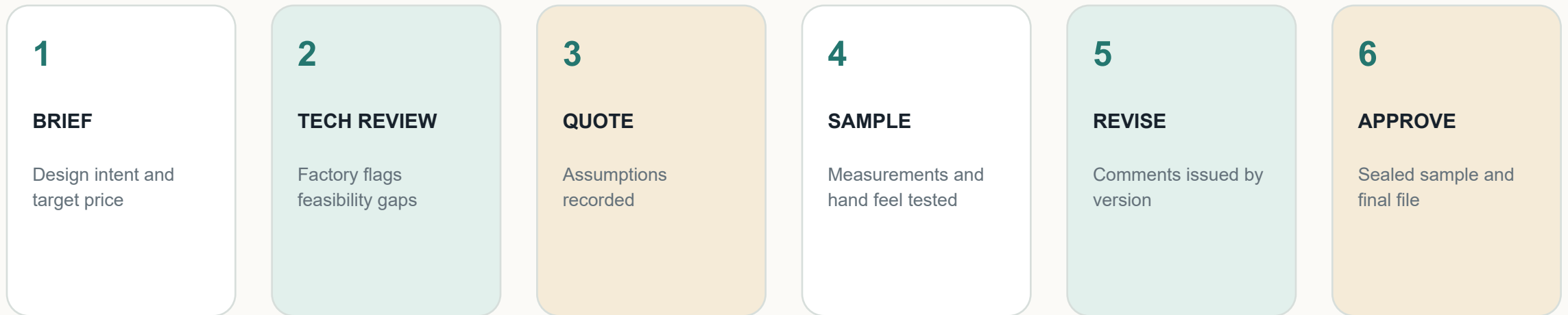
MARKING

PO, style, color, size range and carton number

Record artwork version and placement

A controlled review beats an email handoff

Use a short validation cycle before sampling starts. Resolve commercial and technical gaps while changes are still inexpensive.



Most delays come from missing decisions, not complex designs

Use these failure signals as a pre-submission quality check.

No measurement method	Different teams measure differently	Add diagram and measurement condition
Gauge only	Density and weight remain unknown	Add washed density and GSM/weight
Uncontrolled revisions	Factory uses the wrong comments	Use version number and change log
Subjective hand feel	Approval cannot be repeated	Attach swatch and measurable tests
Packaging deferred	Shipment is delayed at the end	Approve artwork before bulk

Run this checklist before sending the tech pack to the factory

A complete file should support quotation, sampling and bulk production without undocumented assumptions.

☐ Style code, season, revision date and owner

☐ Measurement chart, method and tolerances

☐ Machine gauge and washed stitch density

☐ Labels, packaging and carton instructions

☐ Sample purpose and approval criteria

☐ Front, back and detail drawings

☐ Yarn composition, count, supplier and color

☐ Construction, linking and finishing notes

☐ BOM and colorway matrix

☐ Open questions clearly marked for factory review

Make the next sample easier to approve

Use the knitwear tech pack as a shared decision record. Confirm measurements, yarn, gauge, construction and packaging before the factory starts sampling.

Visit YouTricot Official Website

youtricot.com

Continue your sourcing research

- [Contact our sourcing team](#)
- [Explore more knitwear resources](#)
- [Download this whitepaper from the GitHub repository](#)

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